

Small language modeling with rotary attention and causal suffix copying

DASE7506 / MP1 | Final frozen local experiment | 29 September 2026

1. Result and benchmark protocol

The final predictor achieves 1.5501346122 full-test bits per byte (BPB), a 26.23% reduction from the original baseline's 2.1012651958. It combines an 8,278,080-parameter Transformer trained from random initialization with a stateless exact-suffix copy distribution. The selected validation score is 1.53268101. Training was stopped at update 8,400 when submission preparation began; the aspirational target below 1.5 was not reached.

Measure	Final result	Limit
CPU scoring / baseline median	3.022 x	5 x
Peak evaluation process RSS	1.638 GiB	4 GiB
Uncompressed inference assets	31.707 MiB	64 MiB
Test coverage	428,405 targets /1,292,013 bytes	Complete test
Report length	5 pages including references	10 pages maximum

Protocol7506-mp1-wt2-v2 uses the supplied WikiText-2 raw text and train-fitted BPE tokenizer (vocabulary 2048). The original scorer uses independent causal windows of 256 targets, including the final short window. Every token after the first split token is scored once. BPB equals total negative log2 probability divided by the full split raw UTF-8 byte count. The evaluation mode is CPU FP32 with 4 threads; token perplexity is not the ranking metric.

All learned weights use only the supplied training text. Architecture and checkpoint choices use validation, with no external corpus, pretrained weights, retrieval database, cached validation/test answers, cross-window state or network access during evaluation. The final weights, source and mixture settings were frozen before this round of test scoring. Earlier published test scores are historical evidence, not new-model selection criteria.

2. Neural architecture

The backbone has 6 blocks, width 320,4 attention heads and context 256. Attention is pre-RMSNorm, uses bias-free query/key/value projections, rotary position encoding (RoPE) and a causal attention mask. Adjacent channel pairs are rotated with frequencies $10000^{(-2 \cdot i/d)}$, preserving norms and making query/key dot products depend on relative positions. There is no learned absolute-position table.

A bias-free SwiGLU feed-forward network uses hidden width 896, followed by a down projection. RMSNorm uses epsilon 1e-5. Dropout 0.2 is applied to embeddings and residual branches during training and disabled in evaluation. Input embeddings and the output decoder share weights. Residual output projections use standard deviation 0.02/sqrt(2 L); other linear/embedding weights use 0.02. These choices are a combined architecture/regularization change, not individually isolated claims.

3. Exact within-window suffix copying

For the prediction following input position t , look for prior contexts ending at j whose last k tokens equal the current last k tokens, with $k \in \{1, 2, 3\}$. Only $j < t$ is eligible. Its successor $x[j+1]$ is therefore already observed at or before t . Choose the longest suffix order with at least one match; distribute probability uniformly over the matching occurrences and add their mass to the corresponding successor tokens.

If q is the neural distribution and c is this empirical continuation distribution, the final distribution is $p = (1 - \alpha)q + \alpha c$. The selected α values for suffix lengths 1, 2, 3 are 0.1, 0.3, 0.5. If no suffix matches, $p = q$. These weights were selected in the earlier validation search and held fixed during the new rotary trajectory. Repeated successor tokens accumulate their counts; no future target is passed into `predict_log_probs`.

All match matrices and counts are temporary tensors for the current input call. No state survives a window, batch or scoring pass. The last key is explicitly discarded because its successor is not visible. A shorter or padded sequence cannot influence earlier positions through either the strict copy mask or causal neural attention. Copying adds $O(KT^2)$ matching work and transient $O(T^2 + TV)$ storage per example, for $K=3, T \leq 256, V=2048$.

4. Training and selection

The local Apple M5 run used FP32 MPS training, seed 17, batches of 32 sequences and 8192 targets per update. AdamW used initial rate 0.001, weight decay 0.1, 100-update linear warmup, a 24,000-update cosine schedule with a 10% floor and gradient norm clipping at 1.0. We stopped after 8,400 updates (68,812,800 targets). Keeping the original 24,000-update horizon is essential when reproducing this early-stopped model.

Training was saved in 300-update chunks. Resumption preserves optimizer, CPU/MPS dropout RNG and the separate sampler RNG. CPU FP32 validation ran every 1,200 updates using the unchanged scorer. The development loader opens only train and validation text. CPU validation uses a replica without altering MPS training RNG. Fixed-copy scoring was subsequently run between training processes to avoid CPU/GPU contention.

Update	Neural validation BPB	Fixed-copy validation BPB
1200	1.74161605	1.67271168
2400	1.66640037	Not measured
3600	1.63007361	1.57985715
4800	1.59928902	1.56013117
6000	1.57869232	1.54712196
7200	1.56936363	1.54022325
8400	1.55865998	1.53268101

The final choice is the single update 8,400 checkpoint with the fixed copy setting, the lowest observed validation candidate available when the user requested finalization. No checkpoint averaging or ensemble is used in this release. No weight, hyperparameter or source change was made in response to the final test score.

5. Equal-target comparison and mechanism ablation

Predictor	Training targets	Validation BPB
Original baseline, update 1,200	9,830,400	2.07108743
Rotary neural, update 1,200	9,830,400	1.74161605
Rotary + fixed copying, update 1,200	9,830,400	1.67271168
Final neural weights, copying off	68,812,800	1.55865998
Same final neural weights, copying on	68,812,800	1.53268101

The first two rows control the number of processed training targets. They support the quality improvement of the combined larger rotary/regularized model and its training schedule at equal data exposure. They do not isolate capacity, RoPE, normalization, activation or schedule, and do not equalize compute: the new model has 8.28 million parameters versus 1.09 million in the original baseline. The later final gain also includes longer training.

The final two rows provide a paired ablation with identical neural tensors and training ancestry. Turning on only the suffix-copy mechanism reduces validation BPB by 0.02597897 (about 1.67%). At update 1,200 the same mechanism gains 0.06890437 BPB. Its smaller late-stage benefit is consistent with the stronger neural component already modeling more local repetition; this is an interpretation rather than a separately established causal explanation.

6. Failed alternatives and previous development

An earlier 4-layer, width 192 dropout 0.1 model averaged its 5,400/6,000 snapshots and used the same copy weights. It scored 1.65357322 validation and 1.66816836 test BPB. Its earlier two/three-snapshot average comparison and 8 baseline/12 final copy trials remain recorded in results.json and the immutable previous repository commit. Those weights are not ancestors of the new model.

A learned pointer pilot froze the new model at update 1,200 and trained a 41,281-parameter query/key/gate head for 300 more updates (2,457,600 targets), with key width 64, AdamW rate 0.003, weight decay 0.01, seed 17 and CPU 2 threads. Keys at j were paired only with observed successors $x[j+1], j < t$. It scored 1.67805536 validation BPB, worse than 1.67271168 for the fixed suffix mixture on the same parent, so it was rejected despite its additional training cost. This bounded pilot does not establish that all neural pointer models fail.

A two-expert mixture-of-softmaxes prototype was tested only on synthetic correctness fixtures. It had no benchmark training or resource measurement and was not selected. Synthetic architecture/speed probes and random-model CPU validation probes informed local feasibility; none contributes final weights. These experiments are disclosed without shipping their development code or checkpoints.

7. Frozen test and resource evidence

The frozen final checkpoint was evaluated three times, alternating with the supplied trained baseline in fresh CPU processes. Final BPB was identical across all three runs: 1.5501346122. Baseline scoring seconds were 4.579, 4.361, 4.449; final seconds were 13.445, 12.999, 14.205. The median ratio is 3.022. Peak process RSS includes data loading and model construction, measured with macOS `/usr/bin/time -l`; it is not the evaluator's GPU-allocation field.

8. Training and search costs

The final checkpoint has one direct from-scratch ancestry of 68,812,800 processed training targets. Shared parent snapshots are not counted repeatedly. All previous runs remain project search costs even though they contribute no final weights. Times below are recorded process wall times; some CPU pointer work overlapped MPS training, so their sum is not total elapsed machine time.

Work	Targets / trials	Recorded cost
Original baseline	9,830,400 targets	315.42 s training
Earlier interrupted pilot	5,734,400-6,553,600 targets	>=374.69 s; partial work unknown
Earlier width 192 main run	49,152,000 targets	2,435.11 s training
Corpus smoke check	512 targets	0.1112 s training
Earlier copy selection	8 baseline +12 final trials	140.21 s +71.39 s validation
Earlier snapshot averages	2 candidates, shared ancestry	0.141 s construction
Earlier other validation	5 official +10 main +other calls	30.21 s +85.33 s +29.23 s
Earlier main preparation	10 chunks	46.03 s
New rotary main run	68,812,800 targets	4528.26 s training
New preparation / validation	28 chunks /7 neural scores	149.52 s /119.83 s
Rejected pointer pilot	2,457,600 new targets	536.89 s train;99.94 s val
Fixed-copy saved trajectory	4 scores	55.43 s
Random CPU resource probes	18 validation passes	258.16 s
Trained idle resource controls	7 validation passes	50.79 s
Synthetic throughput probes	63 updates /516,096 targets	No benchmark targets; discarded
Final official test repeats	3 baseline +3 final	54.04 s scoring

The first fixed-copy control on update 1,200 took 97.85 s while GPU training was active; the same prediction took 9.95 s with training stopped. The slower observation is retained as search cost, not used to certify the final resource ratio. Legacy synthetic probe total wall time and unrecorded partial work from the early interrupted run cannot be recovered and are explicitly unknown. Correctness/recovery tests used discarded synthetic fixtures; their exact aggregate duration was not retained. A baseline score completed before sandbox resource collection failed (4.4761 s); it is excluded from the timing pairs.

Detailed curves, raw scalar timing results, prior copy-search trials and checkpoint/source hashes are consolidated in `code/results.json`. The rejected pointer adds only its new targets to total cost; its update 1,200 parent is already counted in the main trajectory. No synthetic or rejected model is an inference dependency.

9. Reproduction and critical analysis

README.md gives exact pinned installation, training and evaluation commands. Direct evaluation uses `python evaluate.py --checkpoint release/checkpoint.pt --device cpu --precision fp32 --threads 4 --split test`. The code archive and checkpoint archive share the `MP1_project/` root and can be extracted together. Evaluation requires no retraining and works offline after installing dependencies. The code repository includes all model imports, benchmark files and tokenizer; the checkpoint contains the final FP32 parameters and configuration, without optimizer state.

For training reproduction, use `configs/scaled.json` with `--steps 24000, --chunk-steps 300, --eval-every 1200, --device mps` and seed 17. Run the initial chunk and 27 resume chunks, then stop. Load `step-08400.pt` and set only its `copy_weights` to `[0.1,0.3,0.5]`. CPU training is also available, but floating-point training across device/library builds need not produce bitwise-identical weights. Direct scoring of the supplied checkpoint is the primary reproducibility claim.

The suffix mixture is useful for repeated local names and phrases, but cannot retrieve words outside the fixed window. A matching string can also precede the wrong continuation, so stronger copy weights can harm prediction. Scalar mixture weights do not adapt to semantic confidence. The neural changes and larger capacity cost more CPU work; this result satisfies the measured limits on the recorded machine, not a claim of identical runtime on every CPU.

Only one main random seed was used. The equal-target comparison bundles architecture, regularization and schedule; separate ablations and multiple seeds would be needed to attribute benefits to individual components or estimate run-to-run uncertainty. Repeated validation selection may overfit the development split. The earlier model had already been tested before this new development round; all choices for this round used train/validation only, and the new final test was evaluated only after its own freeze. Repeated frozen test passes measured reproducibility and resources, not further tuning.

10. Integrity, attribution and AI assistance

The supplied evaluator, common code, baseline, data and tokenizer are unchanged and their hashes are recorded. Fresh review found no Critical/Important causal-inference or training-recovery defects. OpenAI Codex substantially assisted with interpreting requirements, method design, code and tests, local experiments, review, source checking and documentation. The student must personally understand the implementation; this report does not assert that the student has completed that review. Course submission and later public peer review are user actions and have not been performed by the assistant.

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References

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References[5-6] motivate recent-history mixtures; the submitted mechanism uses exact suffix counts and does not reproduce their neural cache/pointer architectures or claim their published word-level scores.